STICS crop model general presentation

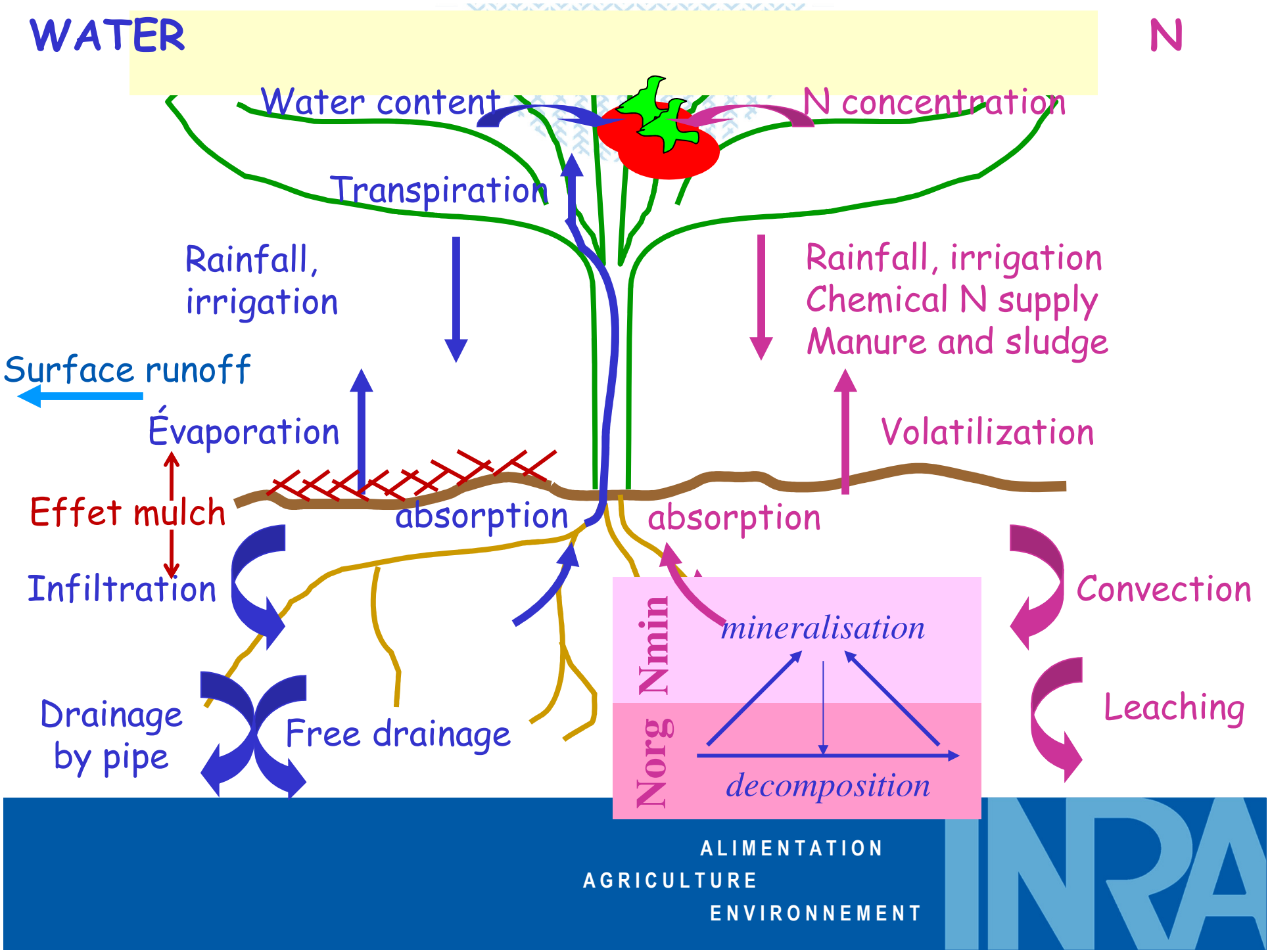
1. Introduction : History, purpose, structure and types of application.
2. Description of soil and formalization of energy, water, Nitrogen balance
3. Formalizations of development and growth of crops
4. Need in inputs and care for options
5. Principles for calibration and test

Description of soil and formalization of energy, water, Nitrogen balance

M Launay, N Brisson, Iñaki Garcia de Cortazar, B Mary, S Andrianarisoa, N Beaudoin; on benhalf of the EPS

ALIMENTATION
AGRICULTURE
ENVIRONNEMENT

INRA


WATER
N
Water content
N concentration
Transpiration
Rainfall,
irrigation
Rainfall, irrigation
Chemical N supply
Manure and sludge
Surface runoff
Évaporation
Volatilization
Effet mulch
absorption
absorption
Infiltration
Convection
Norg Nmin
mineralisation
Leaching
Drainage
by pipe
Free drainage
decomposition
ALIMENTATION
AGRICULTURE
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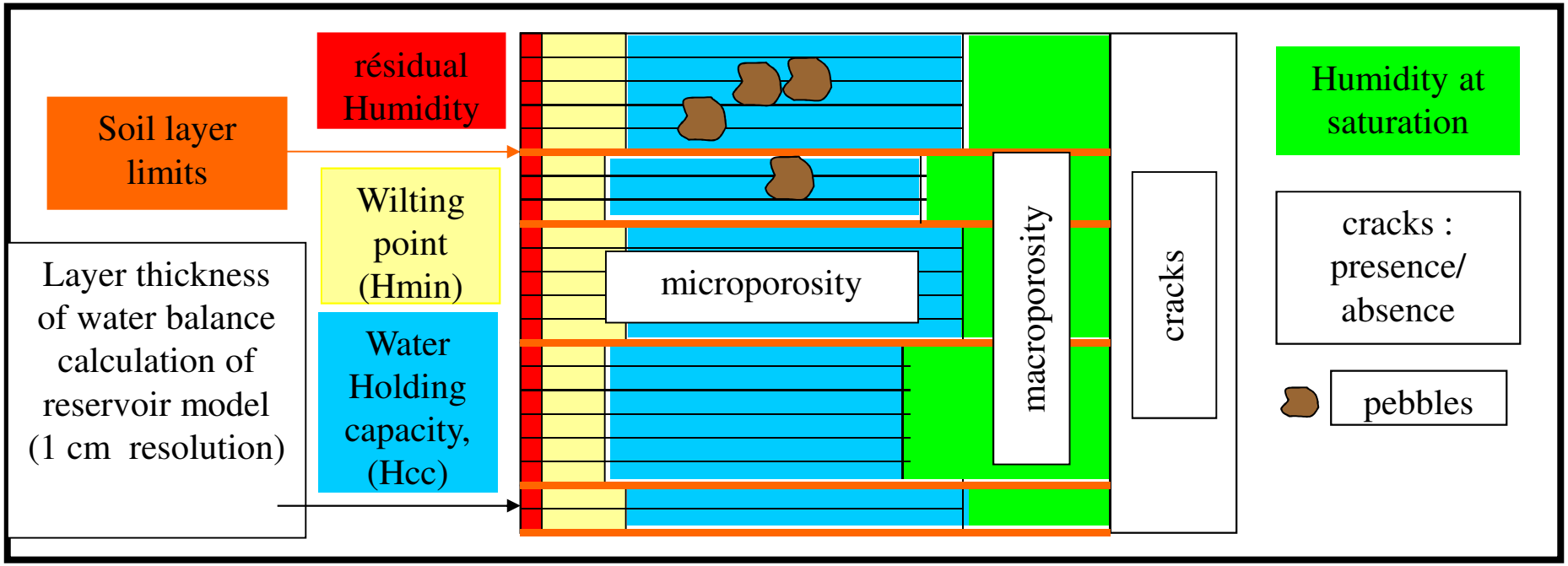
INRA

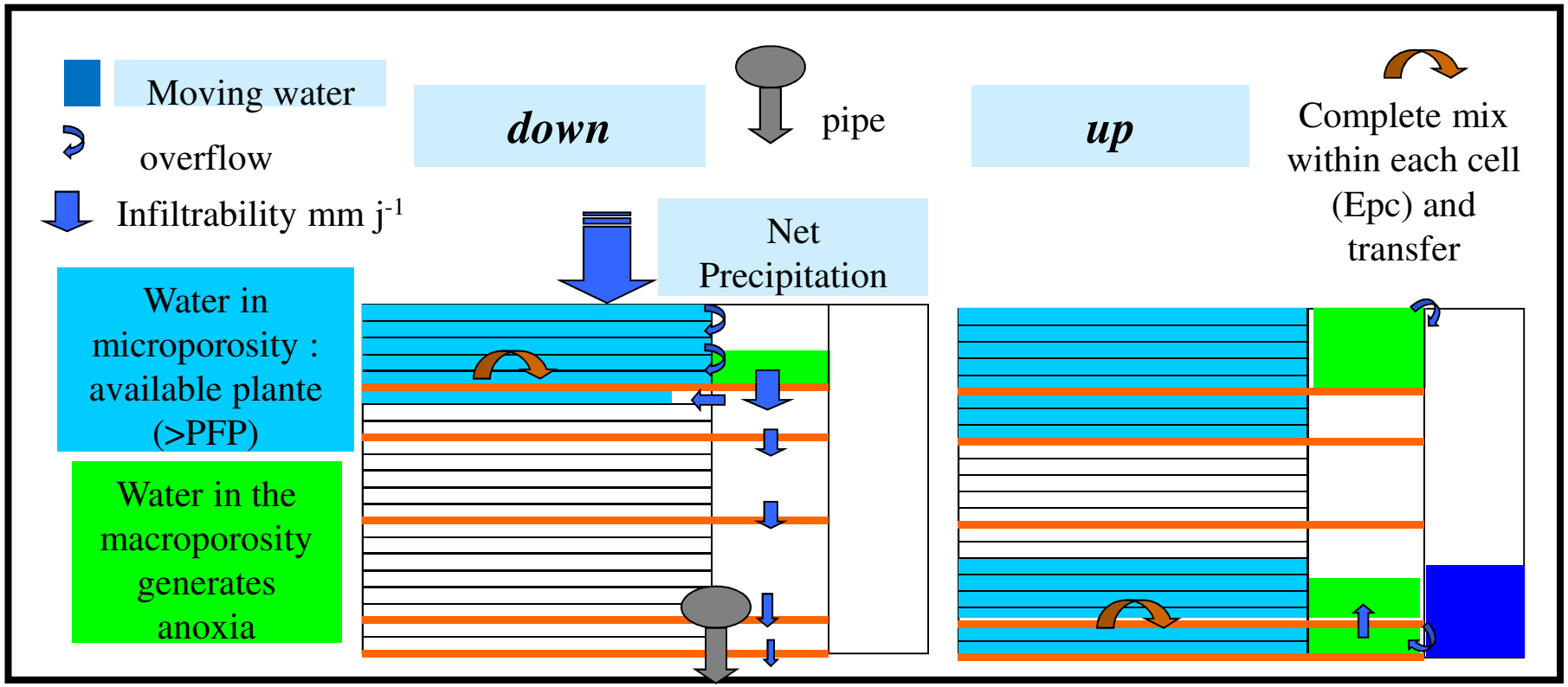
Description of sol compartiment

Upper soil layer => some parameters used for energy, water C, N.

Soil profile = n layers => Permanent features for water and solute storage (agregats and pebbles)



Soil profile = Water flow and soil nitrate transfer in soil



Soil structure evolution

1) Crusting of soil surface according to rainfall

2) Soil structure compaction or fragmentation

Modification structure	Technique	Effets
Compaction	Semis / Récolte (paramètres spécifiques)	Si humidité > seuil (/HCC) ⇒ • ↗DA sur au max 2 couches • ↘ épaisseur de(s) couche(s) • ↘ infiltrabilité de(s) couche(s)
Fragmentation	Travail superficiel / Labour (paramètres spécifiques)	⇒ • ↘DA sur les couches travaillées • ↗ infiltrabilité de(s) couche(s) • ↗ rugosité de(s) couche(s)

Energy balance

Calculations:

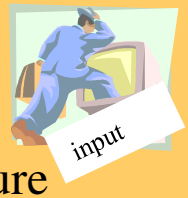
- net radiation
- culture temperature

Calculation of net radiation RNET

RNET = (1-albedo) TRG + RGLO

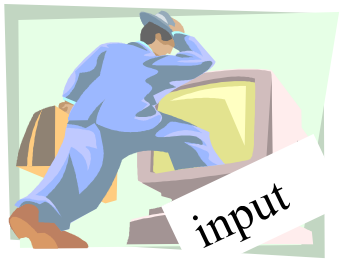
Albedo of soil or canopy

soil
- Dry soil Albédo
- Soil surface moisture



Canopy LAI

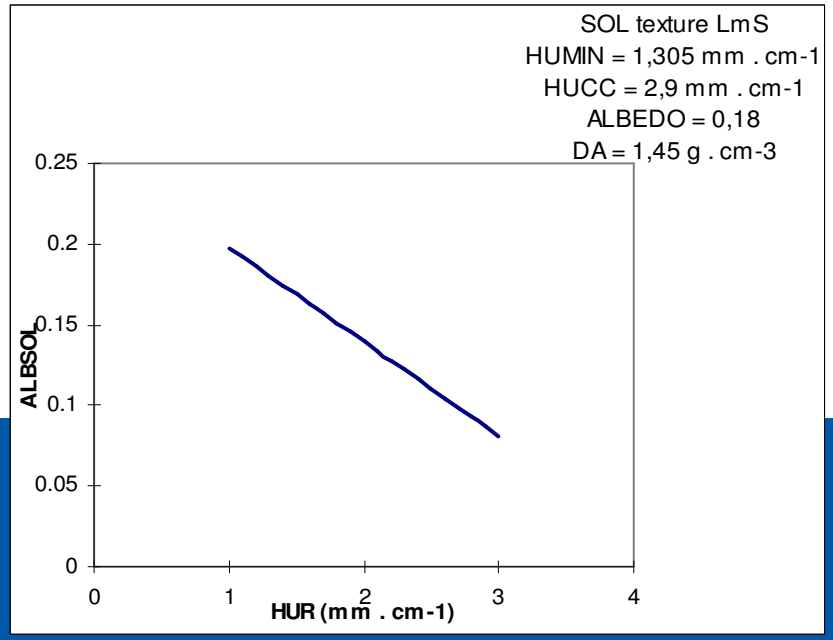
Solar radiation



Big Wavelength λ

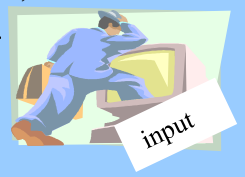
RATM - RSOL

σ Tcult ^{4}



Brunt ou Brutsaert

- Cloudiness (Angström)
- Air Moisture



Tdew = Tmin

Calculation of canopy temperature TCULT

Principle TCULT=(Tcultmin + Tcultmax)/ 2

2 options:

Empirical Relationship
- Tcultmin = Tmin
- Tcultmax = Simplified relatioship (RNET, ET, z0)

Energie balance at Tmin & Tmax

$$RNET = ET + G + H$$

Radiative energy input

$$\rho Cp \frac{TAIR - TCULT}{RATM}$$

Evaporation (ET) + soil temperature increase (G) , air…….. (H)

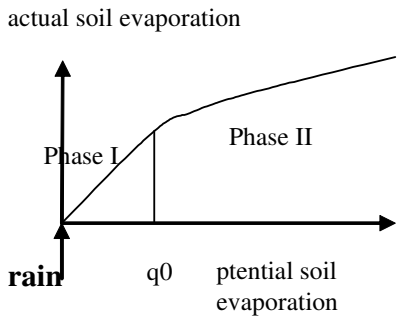
Water balance

Water balance overview

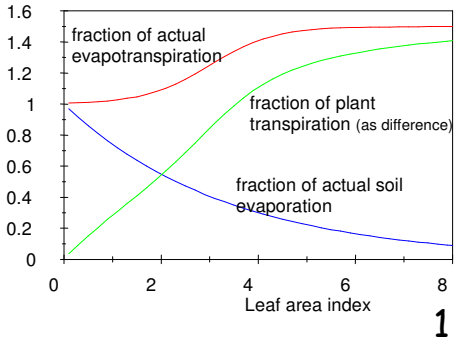
Demand

ETP Penman

soil evaporation



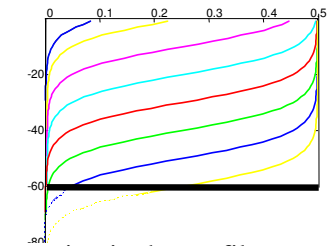
crop transpiration



Supply

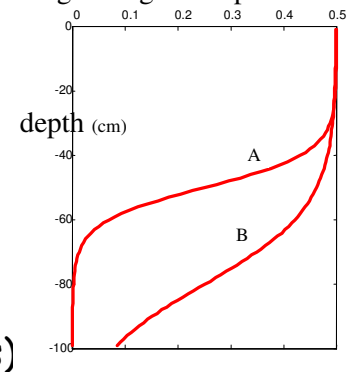
rooting

(growth, final profile)

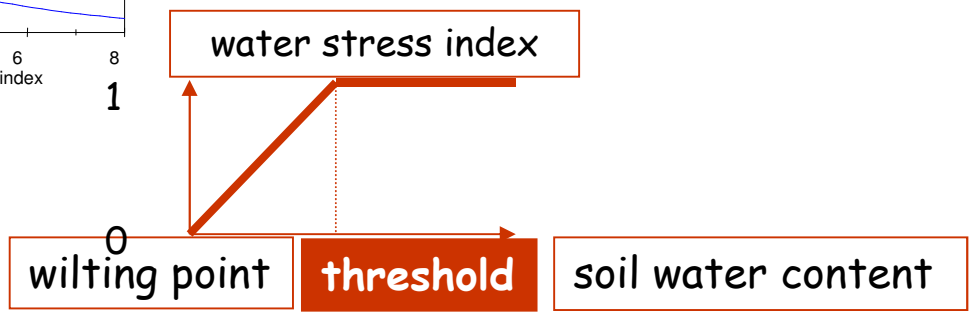


soil water

(depth, water properties)



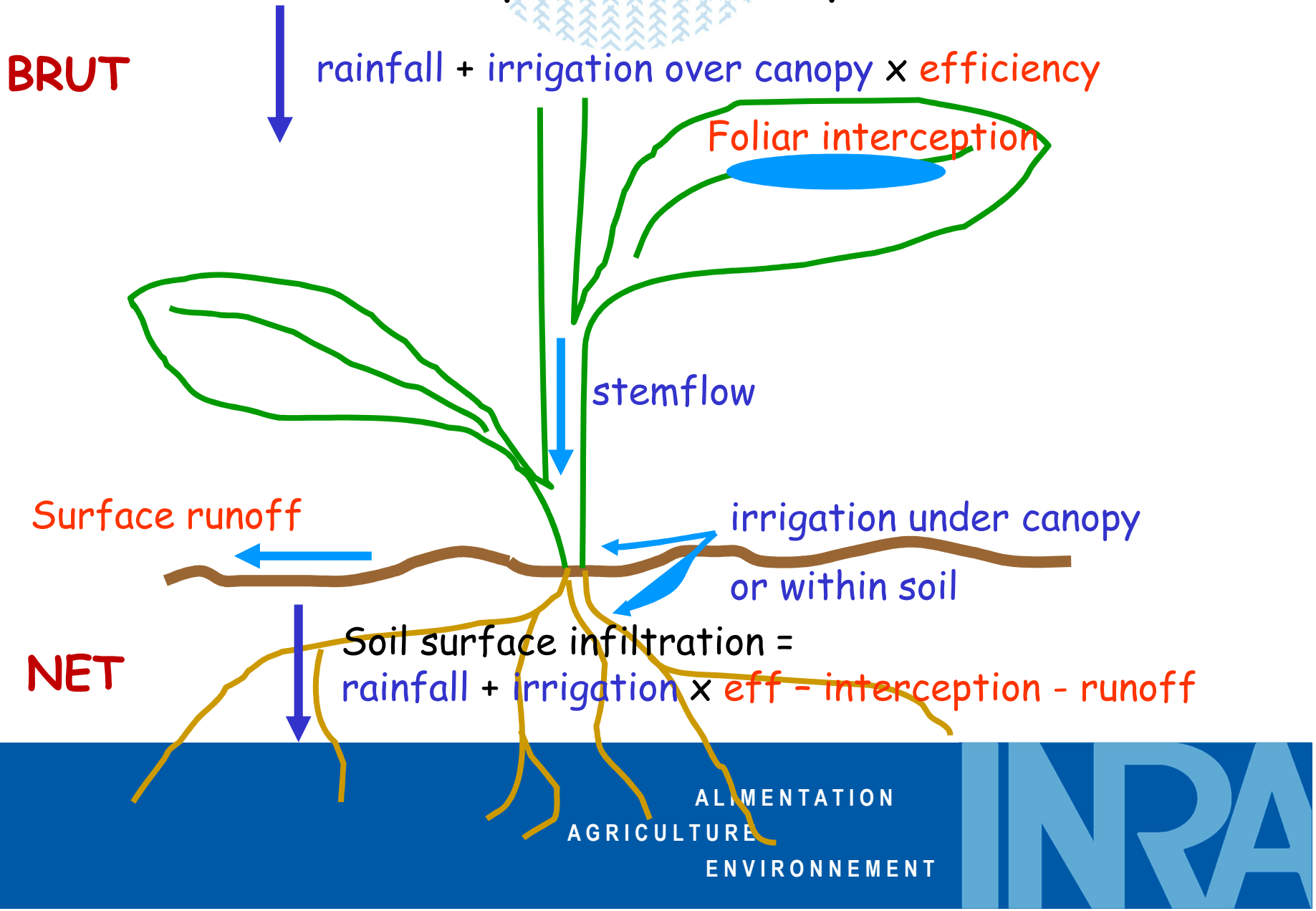
action on physiological fonctions



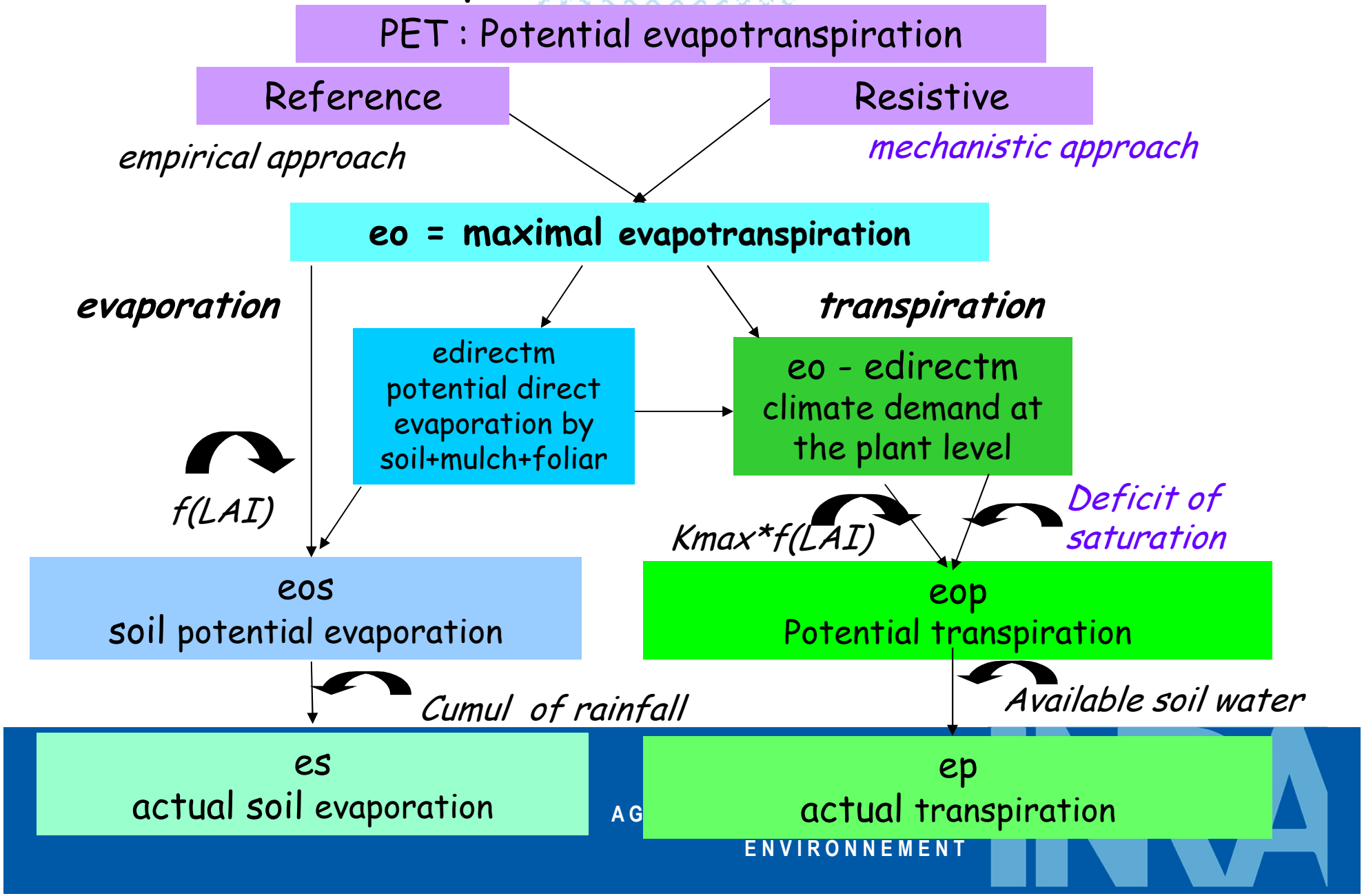
water stress index

Supply/demand

Water inputs of the system



Principles of PET calculation

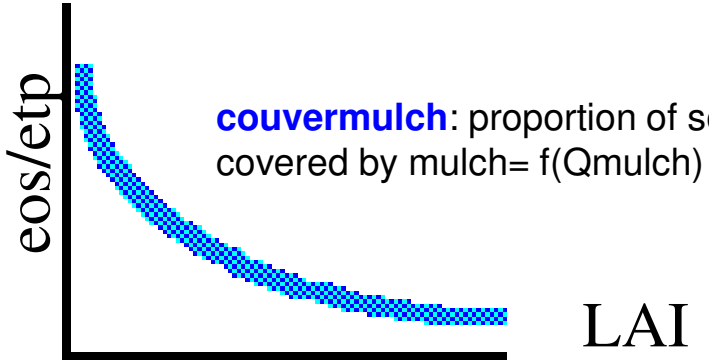


soil water evaporation

1. evaporation **potentielle 'eos'**

Method 1 →

eos = *ETP**(1-couvermulch) $e^{-\delta.LAI}$



couvermulch: proportion of soil covered by mulch= f(Qmulch)

Method 2 → energy balance

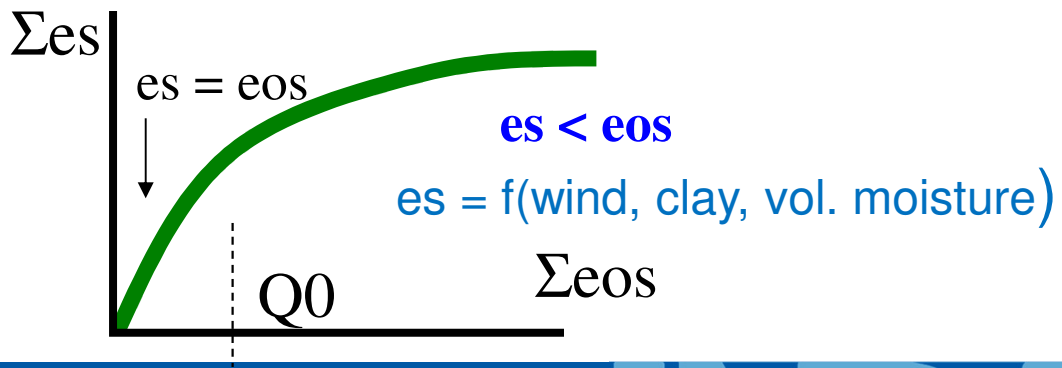
eos = *f(temperature air, rnets, ras, raa, L, GAMMA, tvar)*

L: chaleur latente de vaporisation, **rnets** = rayonnement net au niveau du sol, **ras** = résistance aérodynamique entre le sol et la végétation, **raa :** résistance aérodynamique entre la végétation et le niveau de référence, **GAMMA** = constante psychrometric, **tvar** = pression de vapeur saturante, **dsat** = déficite de saturation de l'air

2. evaporation **réelle 'es'**

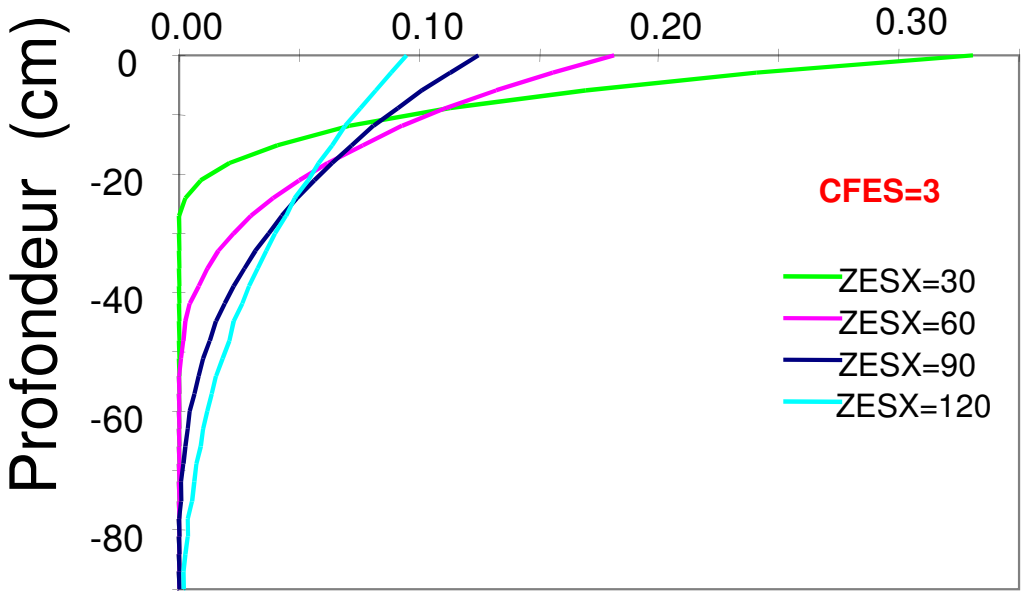
2 successive phases

(Ritchie 1972)



3. Distribution of water deficit due to evaporation within soil profile

Empirical relationship describing contribution of soil water to evaporation according to depth (esz)



esz = f(clay, humidity, cfes, zesx)

CFES: parameter describing the contribution of the soil by evaporation depending on the depth, zesx = maximum depth soil contributing to the evaporation

Water absorption ⇔ transpiration

1. Potential transpiration of crops 'eop'

= 2 approches

Empirical
K x ETP

K=f(LAI)

K=f(covering rate)

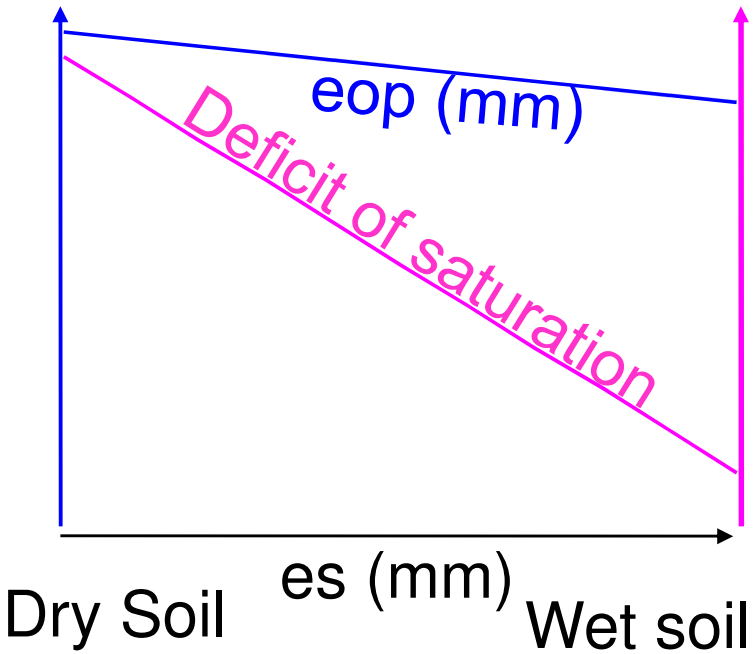
Water Interception
- Canopy
- Mulch

Resistive
(convective,
radiative)

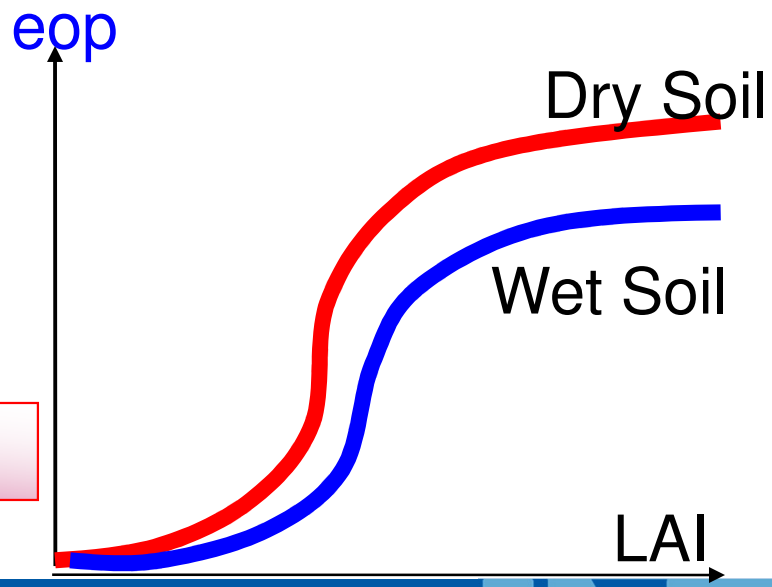
Rnet calculation
- Brunt
- Brutsaert

transpiration ⇔ absorption

1. **Potential** transpiration of crops '**eop** empirical approach



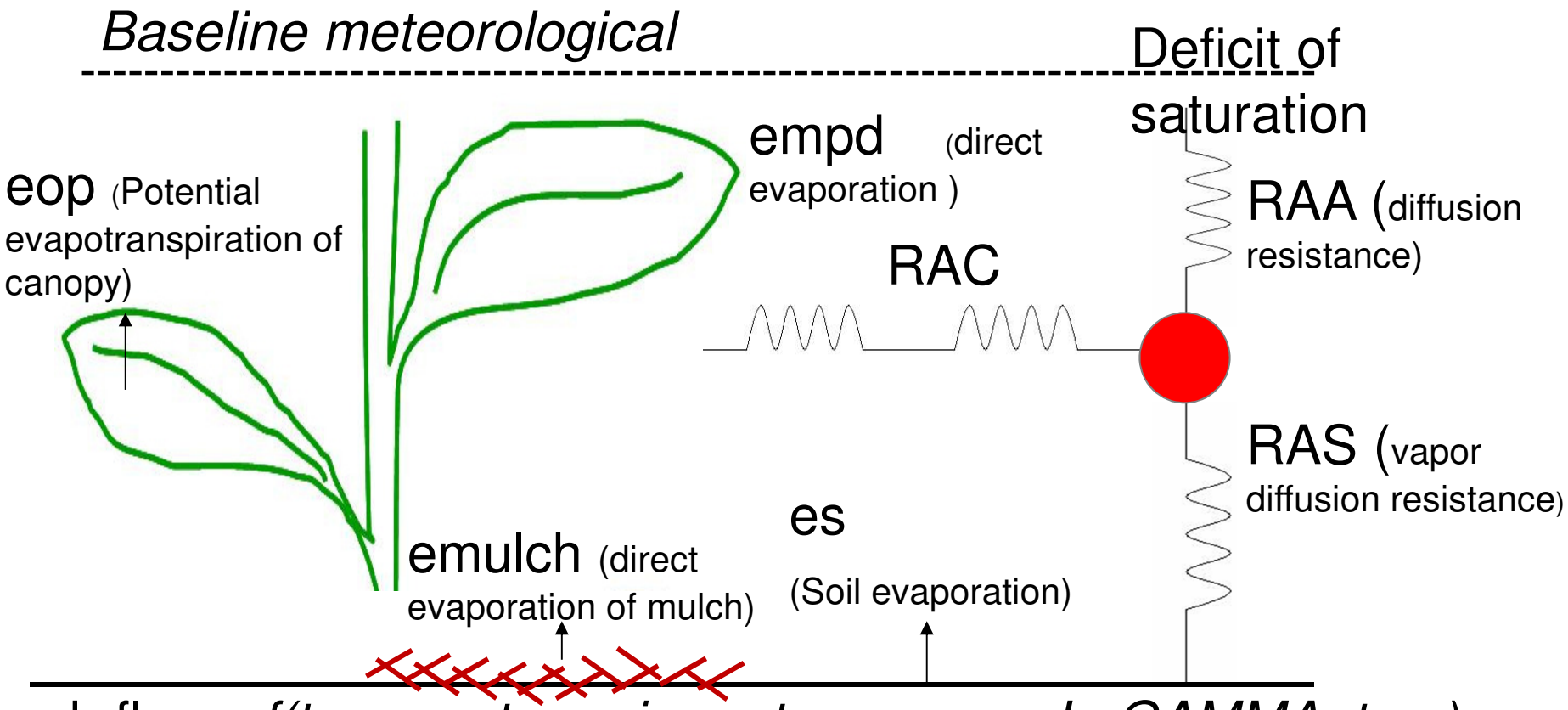
$$eop = \underbrace{(eo - edirectm)}_{\text{Climate demande at canopy}} \underbrace{(\beta - (\beta - 1))\frac{edirect}{edirectm}}_{\text{Saturation deficit (atmospher state under canopy)}}$$



The drier soil the higher eop

transpiration ⇔ absorption

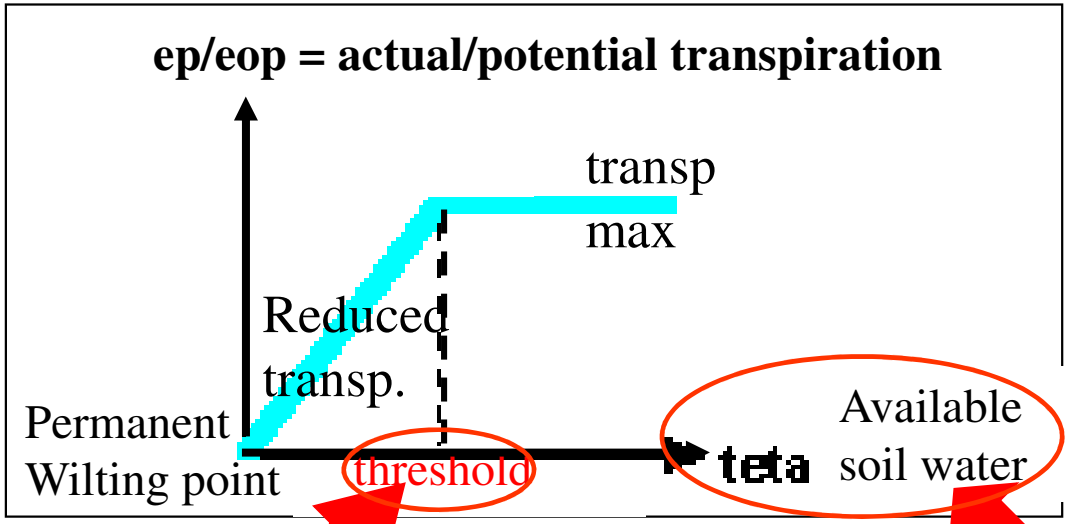
1. **Potential** transpiration of crops **'eop** : resistive approach

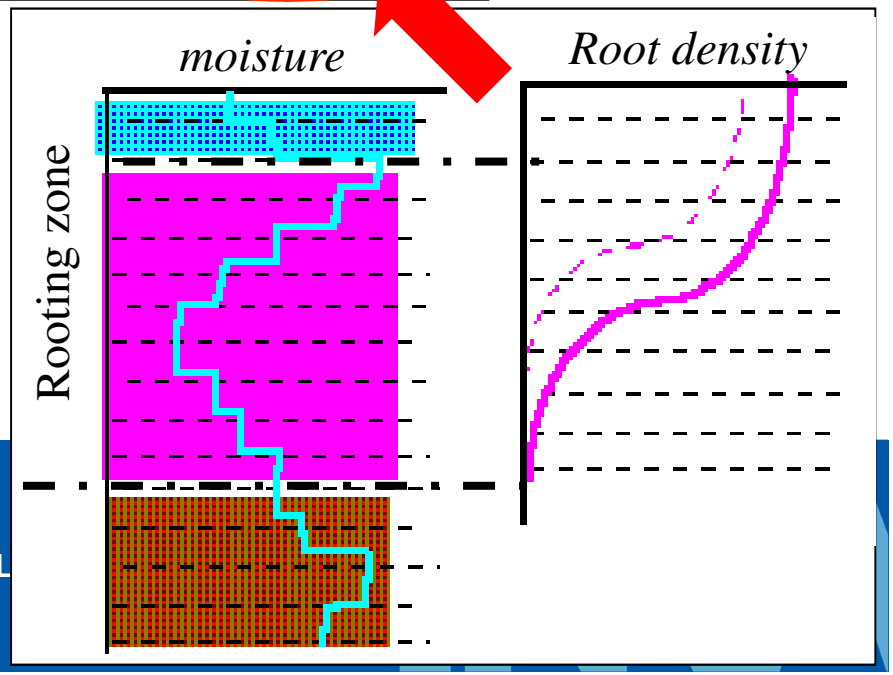


each flux = *f(temperature air, rnets, ras, raa, L, GAMMA, tvar)*

Absorption ⇔ transpiration

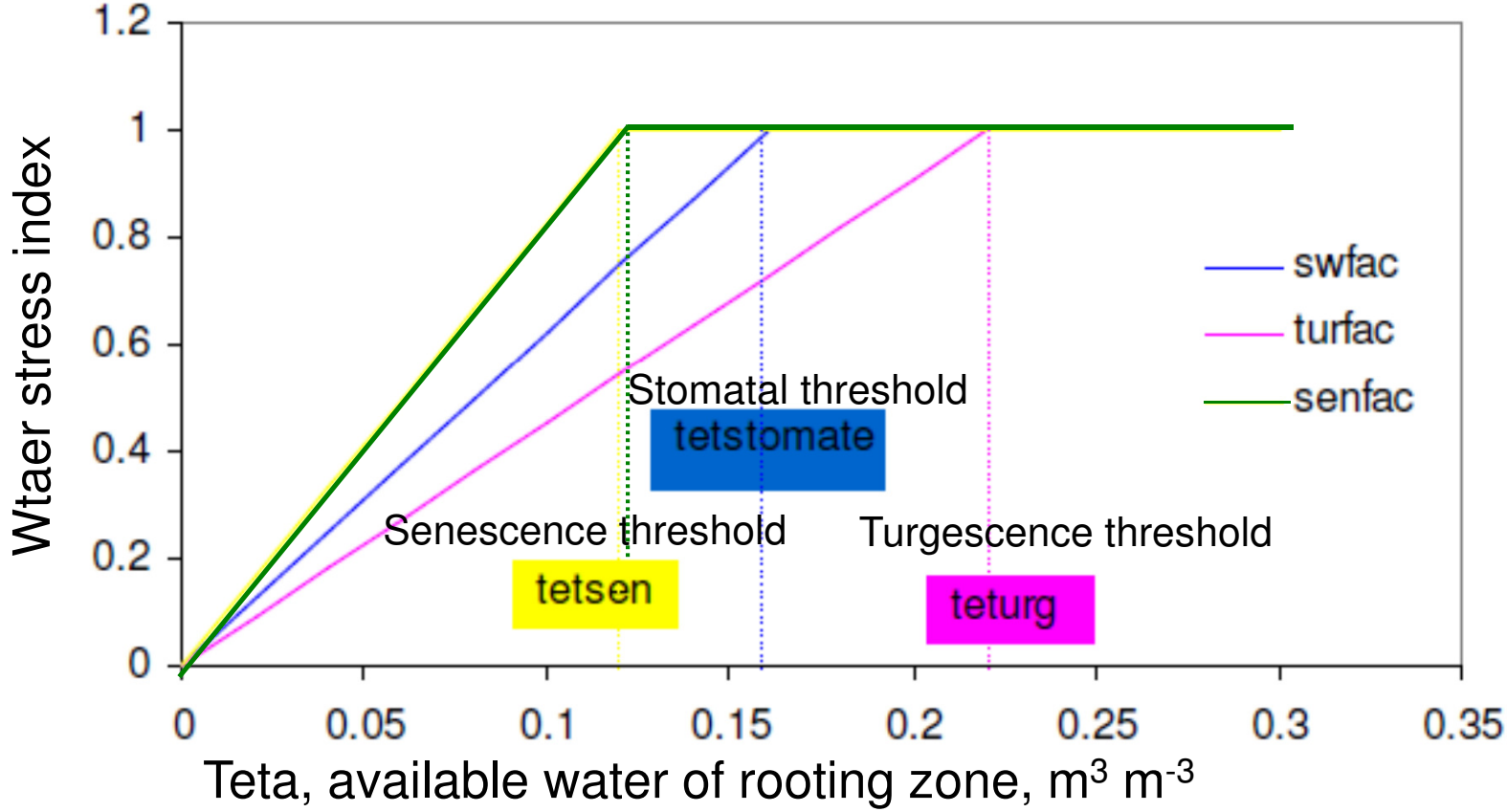
2. Actual transpiration of canopy 'ep'





threshold = f (climate demand, rooting depth, root distribution, potential stomatal closure)

water stress index



SWFAC = stomatal water stress index **transpiration, photosynthesis**

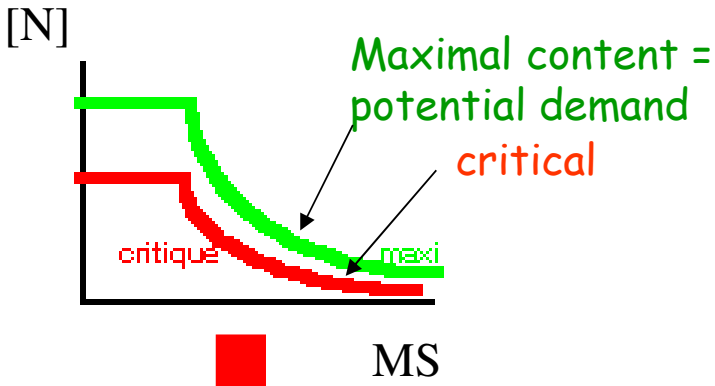
TURFAC = tugor water stress index **foliar growth**

Nitrogen balance

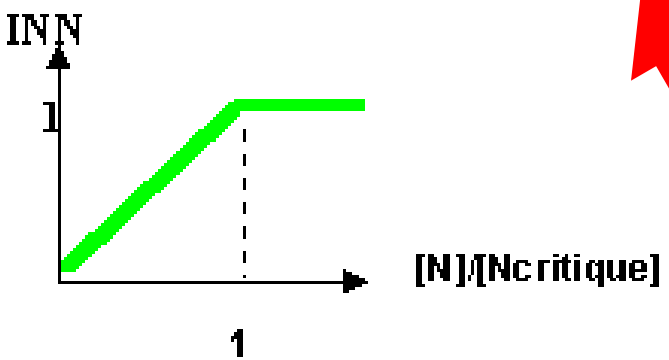
N absorption and Nutrition Index (NNI)

demand

N crop need

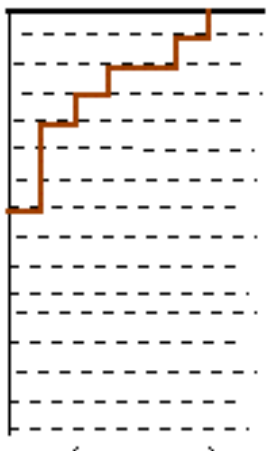


abso = minimum (N need, N supply)



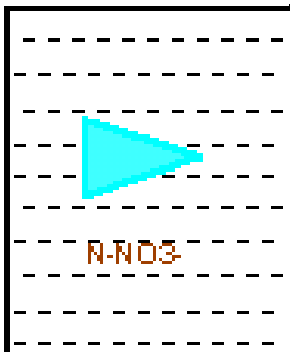
supply

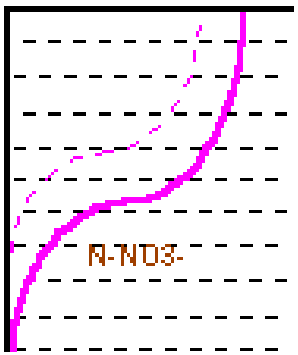
Fertliizer, fixation, mineralisation



quantity
[N, NO_3]

availability
[N, NO_3]

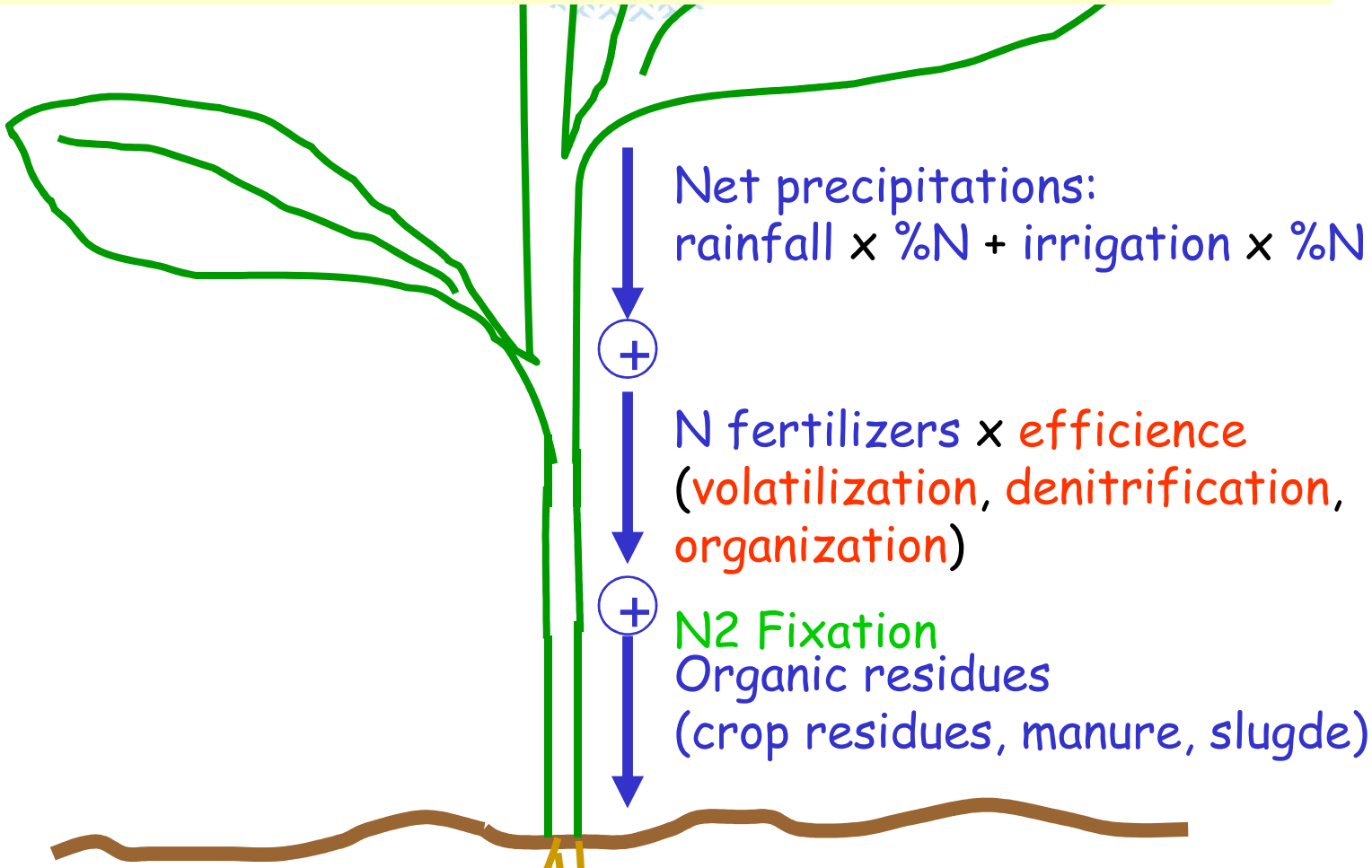




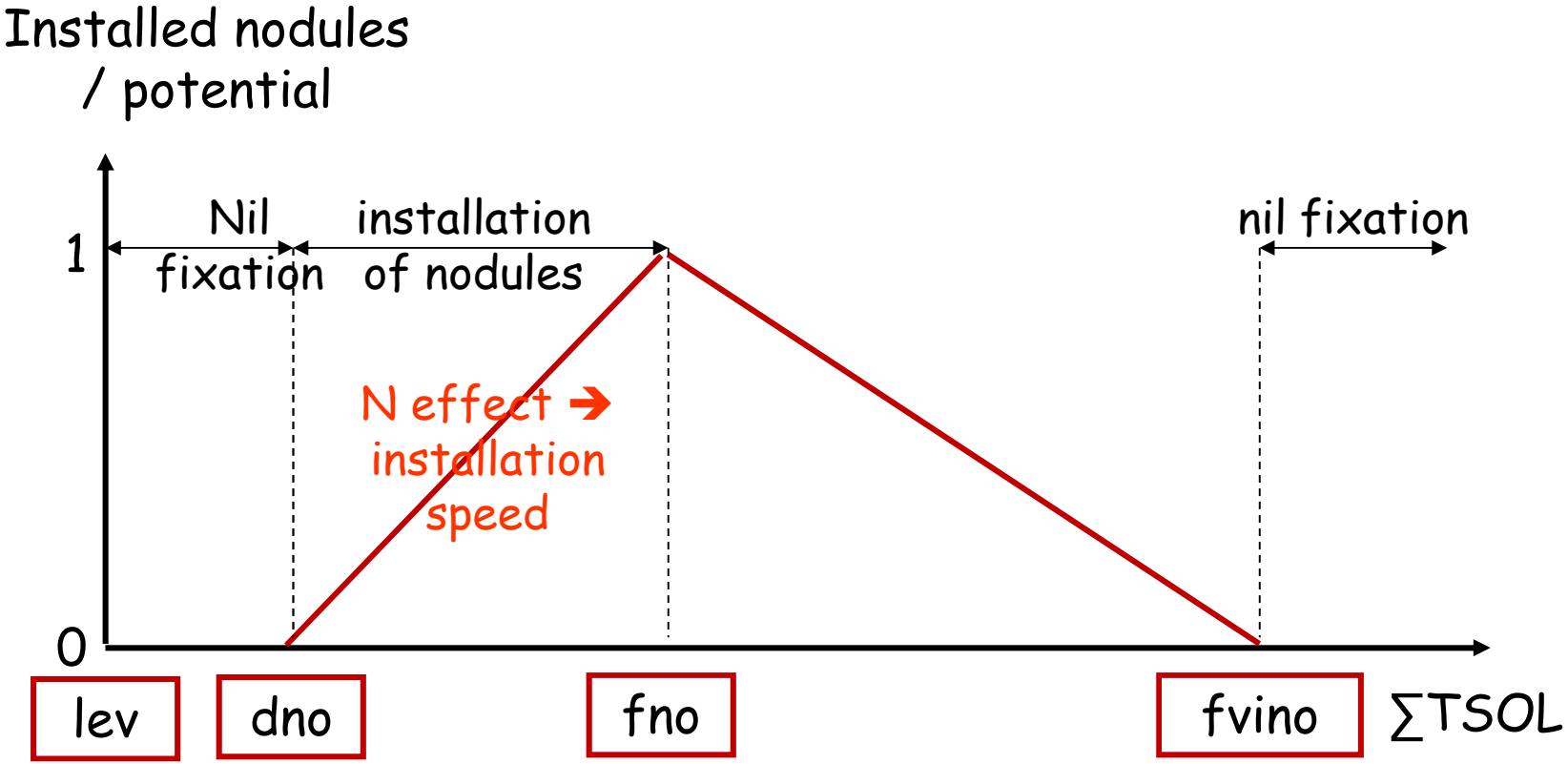
migration absorption active

supply = minimum

Soil-crop nitrogen inputs

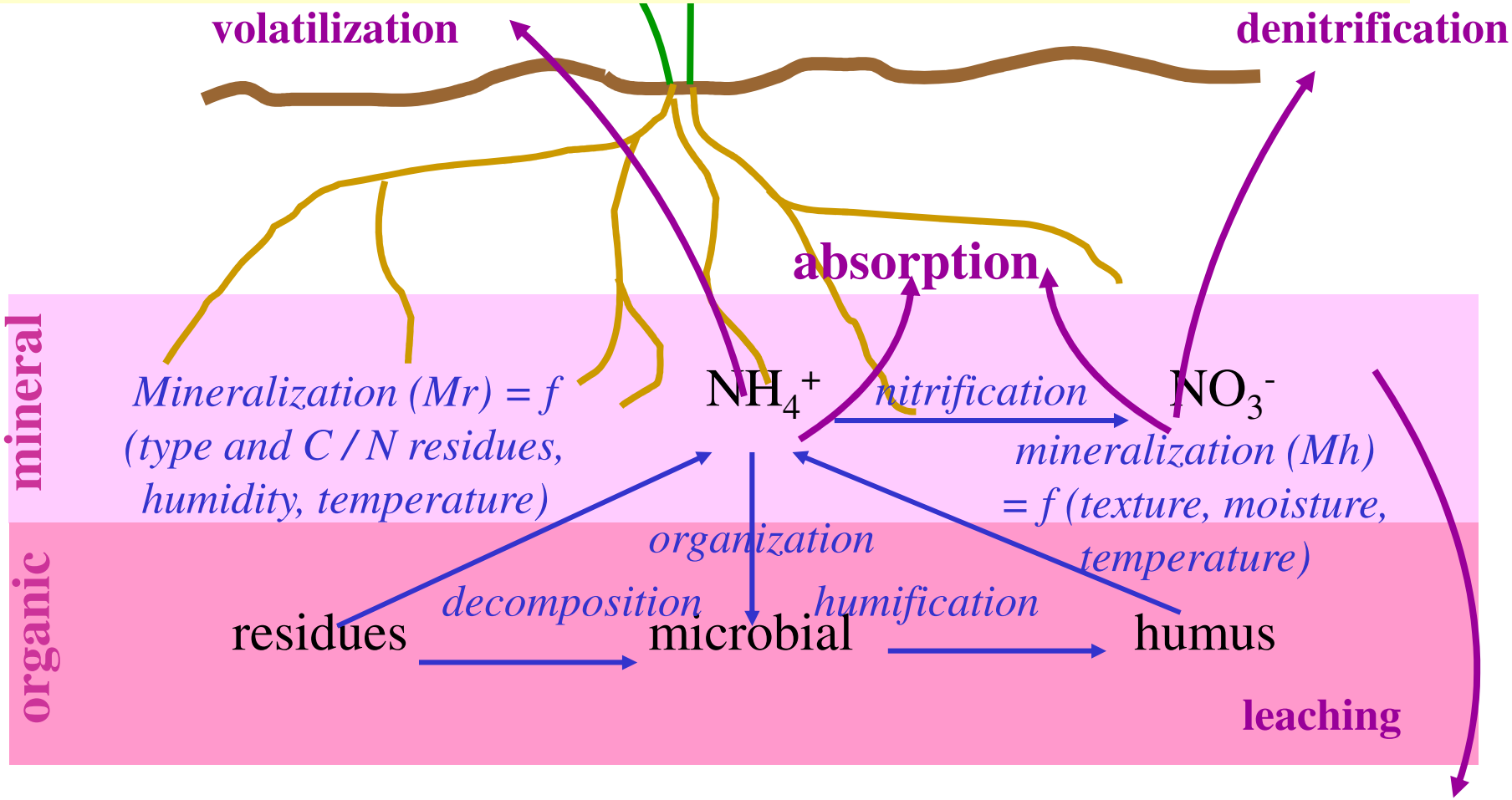


N2 symbiotique fixation

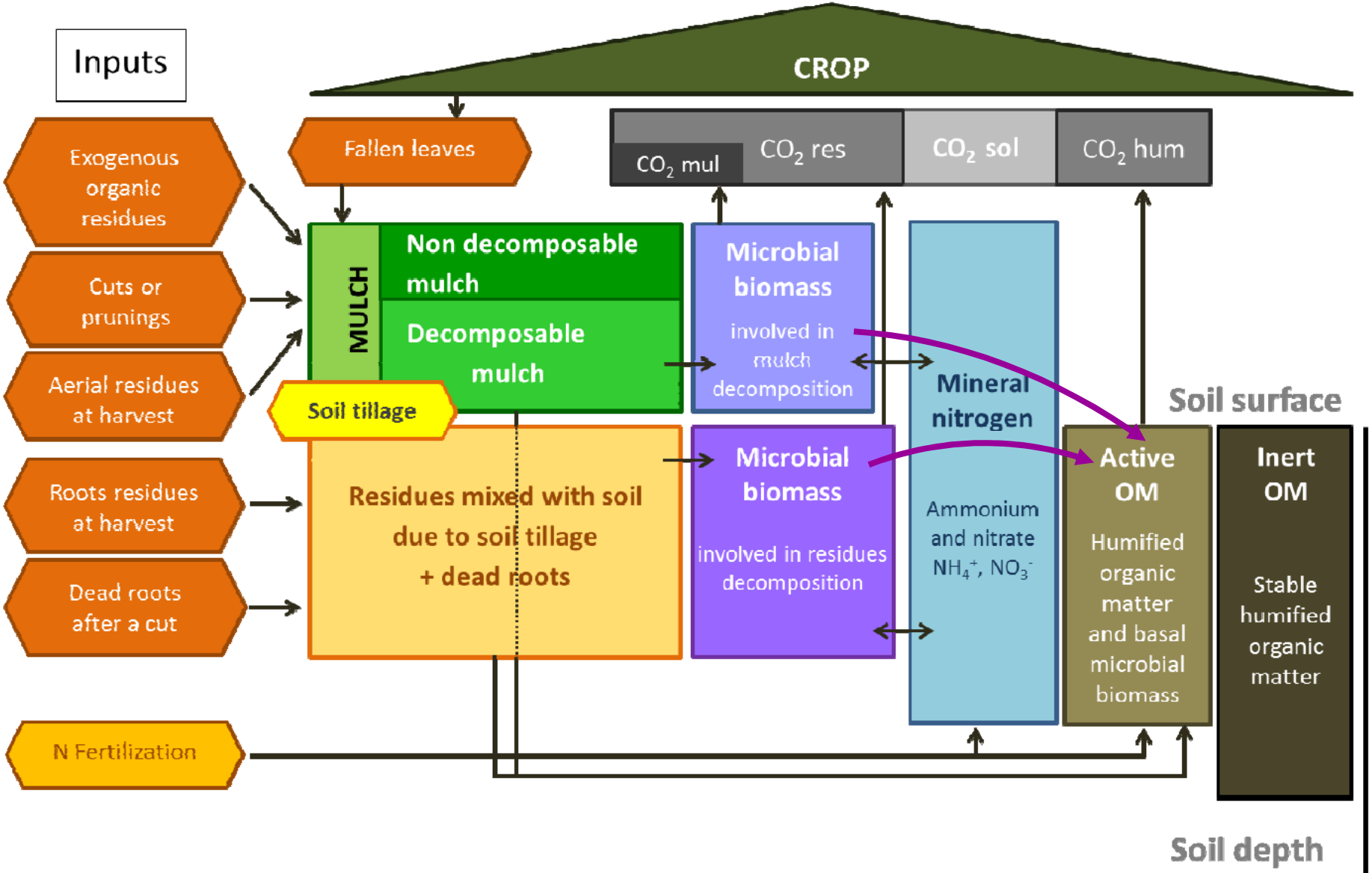


N2 fixation = f (N demande of plante, nodule proportion,
soil mineral N, soil H2O, soil anoxia, temperature)

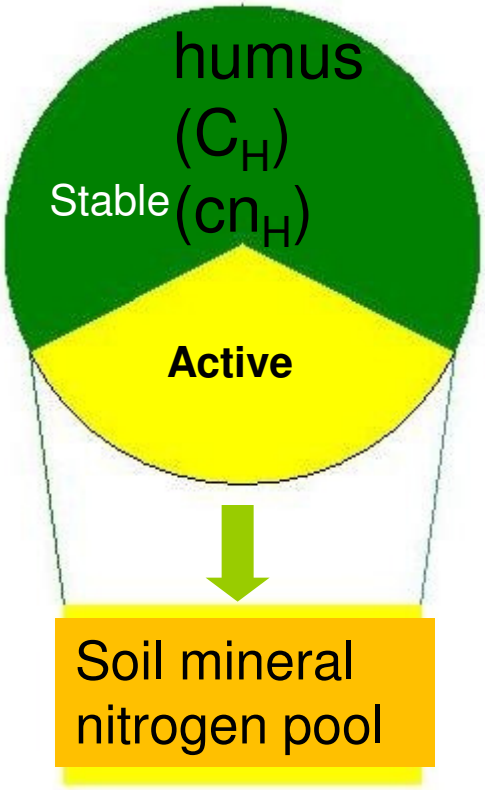
Evolution of the stock of available nitrogen in the soil



Soil C & N compartments and main incoming and outgoing N fluxes (inputs and organic mineralization)



humus mineralisation



stable/total = 0.65

(Andriulo et al., 1999)

Mineralisation rate = VPOT*FTH*FH

Normalized days

$$FTH = \exp\left(FTEM\ (T - T_{REF})\right)$$

$$FH = FHUM + (1 - FHUM)\frac{HUR - HUMIN}{HUCC - HUMIN}$$

- **VPOT** = K2POT*NHUM
- **K2POT** = $\dfrac{0.21}{(12.5 + ARGI)\ (54.5 + CALC)}$

(Not very predictive for sandy soils and non-calcareous ; cf Valé, 2006)

- **NHUM** = Norg*Da*PROFHUM*(1-FINERT)

kgN.ha^{-1} %

Profhum : around plowed layer or highly dense rooting zone

Several types of residues

1) Crop residues or organic wastes left as **mulch or buried** into the soil and biomass associated with microbial residue decomposition (see next slide)

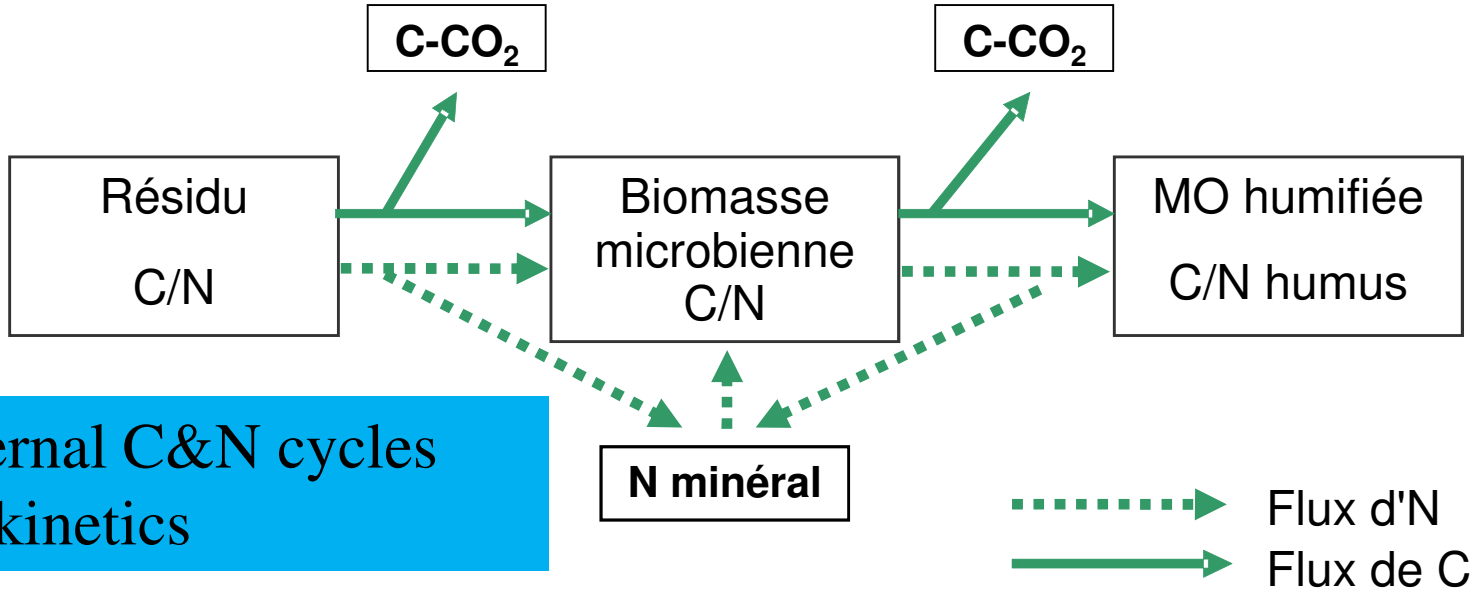
2) Aerial entries of crop residues due to **harvest** ("ressuite" code)

	ressuite	Qressuite	QNressuite
1	'whole_crop'	masec	QNplante-Nrac
2	'straw+roots'	mspaille	QNveget-Nrac
3	'stubble+roots'	mspaille*0.35	(QNveget-Nrac)*0.35
4	'prunings'	mabois	mabois *5
5	'stubble_of_residu_type_9+roots'	mspaille	QNveget-Nrac
6	'stubble_of_residu_type_10+roots'	mspaille	QNveget-Nrac

3) Contributions **of root residues at harvest** or cut =f(root system description option; root length, specific parameter)

Decomposition and mineralisation coupling C&N cycles but regulation

Decomposition = f(Soil moisture and temperature (Normalized days), residues nature and C/N, available soil mineral N (4 levels of regulation)



Soil internal C&N cycles order 1 kinetics

(Nicolardot, 2001),

Successive regulation due N availability: 1: Reduction of the rate of decomposition; 2 reduction of N / C ratio of the microbial biomass; 3: Reduction of N / C ratio of the newly formed humus; 4 emergence of a "Priming effect"(Giacomini et al, 2008).

N Fertilisation

Model Input

Rate, type and date

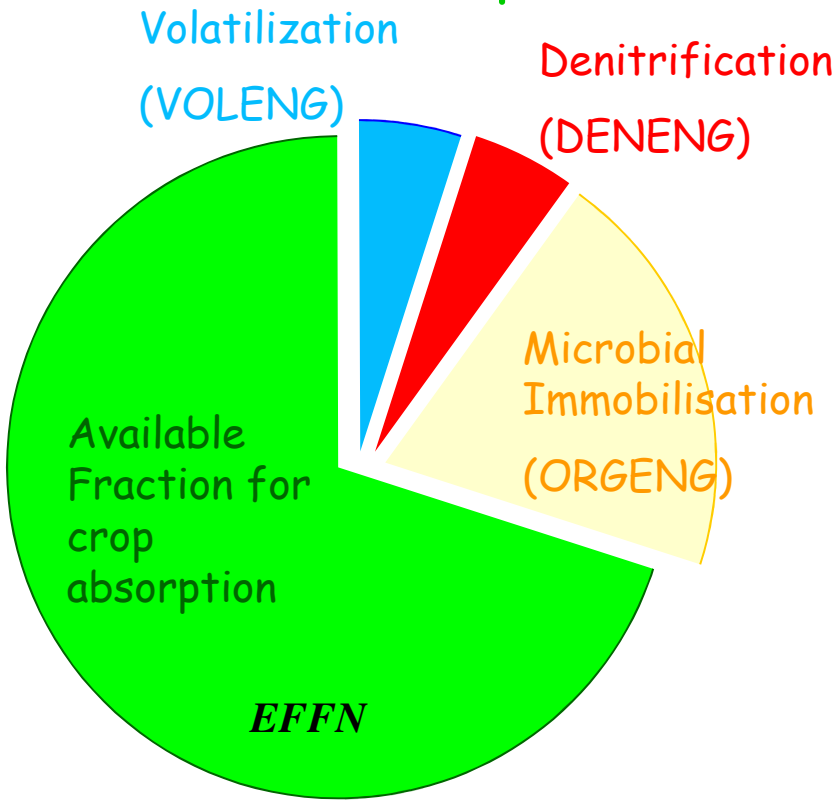
CODLOCFERTI = Localisation code of fertilisation:
1 = at surface of sol; **2** = within the sol

LOCFERTI = Depth of

ANIT = amount of fertilzer

Empirical approach of efficiency and losses

Default option



$$EFFN = 1 - \frac{NORGENG + NVOLENG + NDENENG}{ANIT}$$

In option Nitrification, denitrification and soil N_2O emission (NOE model, Hénaut et al., 2005)

Dénitrification

$$vD = Dp \cdot f_{NO_3} \cdot f_{WFPS} \cdot f_T$$

-Dp (= potential of dénitrification

-f_{NO3}, f_{WFPS} f_T

= response to soil nitrate concentration, porosity saturation and soil & temperature.

Nitrification

$$vN = Np \cdot f_{NH_4} \cdot f_{WFPS} \cdot f_T$$

-Np = potential of nitrification.

- f_{NO3}, f_{WFPS}, f_T

= response to ammonium concentration, porosity saturation and soil & temperature.

- nitrification = min(vN, vMin)

=> possible NH_4 accumulation

Proportion de N_2O

$$vN_2O = r \cdot vD + z \cdot vN$$

- denitrification of NO_3 originating from nitrification

➔ Separate processes in space and time

- r et z = constants (nowadays)

In option : Mechanistic calculation of NH3 volatilization

Purely physical process. The current STICS version only simulates explicitly the volatilization following an application of liquid organic manure (+ empirical calculation for fertilizers). Modelling the balance between four forms: NH4 adsorbed and dissolved forms and dissolved and gaseous NH3

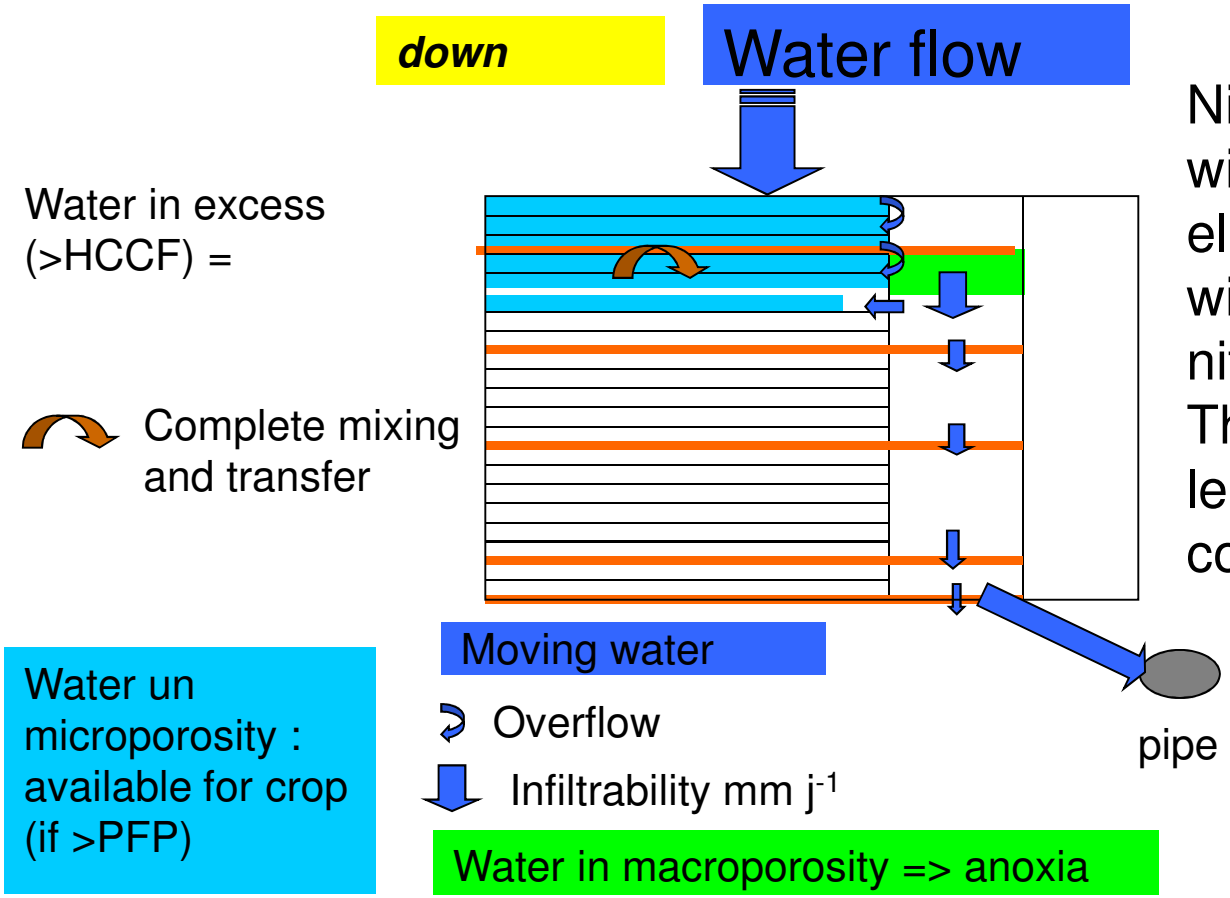
Step 1: Calculate the volatilizable fraction after application, depending on the water content, N min product and tillage.

Step 2: calculation of soil pH after application.

Step 3: Calculation of final proportions of four forms in equilibrium, depending on the pH, temperature, water content and soil porosity.

NO3 Leaching (by percolation or artificial drainage)

= nitrate tranfered under the deepest layer



Nitrate arriving by convection with the water in an elementary layer is mixed with the already present nitrate.
The excess water then leaves with the new concentration of the mixture

The Modulo-Stics version

Thank you for your attention

Thanks to google traduction